

OpenClaw Deployment

Note: It is recommended to refer to the Feishu configuration for the installation method. Downloading openclaw from RDK Studio is relatively simple and easy to use.

1. OpenClaw Installation

One-click installation:

```
curl -fSSL https://openclaw.ai/install.sh | bash -s -- --install-method git
```

To elaborate, this script performs the following operations:

Install corresponding dependencies:

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
312 packages can be upgraded. Run 'apt list --upgradable' to see them.
2026-02-06 17:22:23 - Repository configured successfully.
2026-02-06 17:22:23 - To install Node.js, run: apt install nodejs -y
2026-02-06 17:22:23 - You can use N|solid Runtime as a node.js alternative
2026-02-06 17:22:23 - To install N|solid Runtime, run: apt install nsolid -y
```

Note that GitHub repository needs to be cloned at this step:

Prepare a scientific internet tool that can clone git repositories, otherwise this step may fail

```
✓ Node.js installed
→ Installing OpenClaw from GitHub (https://github.com/openclaw/openclaw.git)...
✓ Git already installed
→ Installing pnpm via Corepack...
Preparing pnpm@10 for immediate activation...
✓ pnpm installed
Cloning into '/root/openclaw'...
```

```
✓ Detected: linux
✓ Node.js v22.22.0 found
→ Installing OpenClaw from GitHub (https://github.com/openclaw/openclaw.git)...
✓ Git already installed
Cloning into '/root/openclaw'...
remote: Enumerating objects: 124672, done.
remote: Counting objects: 100% (1942/1942), done.
remote: Compressing objects: 100% (104/104), done.
remote: Total 124672 (delta 1846), reused 1838 (delta 1838), pack-reused 122730 (from 1
Receiving objects: 100% (124672/124672), 175.79 MiB | 2.53 MiB/s, done.
Resolving deltas: 100% (78113/78113), done.
Updating files: 100% (5025/5025), done.
Already up to date.
```

After installation is complete, you will see the following screenshot:

```
[3/3] Finalizing setup
· Running doctor to migrate settings
✓ Doctor complete

[01F 09E] OpenClaw installed successfully (OpenClaw 2026.3.11 (f2e28fc))!
cracks claws Alright, what are we building?

Source install details
· Checkout: /home/sunrise/openclaw
· Wrapper: /home/sunrise/.local/bin/openclaw
· Update command: openclaw update --restart
· Switch to npm: curl -fsSL --proto '=https' --tlsv1.2 https://openclaw.ai/install.sh | bash -s -- --install-method npm

[01F 09E] OpenClaw 2026.3.11 (f2e28fc)
Claws out, commit in-let's ship something mildly responsible.

Dashboard URL: http://127.0.0.1:18789/
Copy to clipboard unavailable.
Opened in your browser. Keep that tab to control OpenClaw.

FAQ: https://docs.openclaw.ai/start/faq
sunrise@ubuntu:~$
```

After installation, add the environment variable to `~/.bashrc`. The specific path depends on where you installed the files:

```
echo 'export
PATH="/root/.local/bin:/usr/local/bin:/usr/sbin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"' >>
~/.bashrc
or
echo 'export PATH="$PATH:/home/sunrise/.npm-global/bin"'>> ~/.bashrc
```

Let's refresh and configure the environment variables:

```
source ~/.bashrc
```

Now you can start using openclaw via command line.

2. OpenClaw Initial Configuration

Use the following command for initial configuration:

```
openclaw onboard
```

Let's configure the options. You can follow the green circles for corresponding configuration:

```
root@ubuntu:~# openclaw onboard
🔒 OpenClaw 2026.2.4 (c75275f) – If it's repetitive, I'll automate it; if it's hard, I'll bring jokes and a rollback plan

OPENCLAW
🔒 OPENCLAW 🔒

OpenClaw onboarding
• Security
  Security warning – please read.

  OpenClaw is a hobby project and still in beta. Expect sharp edges.
  This bot can read files and run actions if tools are enabled.
  A bad prompt can trick it into doing unsafe things.

  If you're not comfortable with basic security and access control, don't run OpenClaw.
  Ask someone experienced to help before enabling tools or exposing it to the internet.

  Recommended baseline:
  - Pairing/allowlists + mention gating.
  - Sandbox + least-privilege tools.
  - Keep secrets out of the agent's reachable filesystem.
  - Use the strongest available model for any bot with tools or untrusted inboxes.

  Run regularly:
  openclaw security audit --deep
  openclaw security audit --fix

  Must read: https://docs.openclaw.ai/gateway/security

• I understand this is powerful and inherently risky. Continue?
  • Yes / ○ No

• Onboarding mode
  • QuickStart (Configure details later via openclaw configure.)
  ○ Manual
```

For this step, use your own API key for the relevant configuration:

```
• Model/auth provider
  • xAI (Grok) (API key)
  ○ OpenAI
  ○ Anthropic
  ○ MiniMax
  ○ Moonshot AI (Kimi K2.5)
  ○ Google
  ○ OpenRouter
  ○ Qwen
  ○ Z.AI (GLM 4.7)
  ○ Copilot
  ○ Vercel AI Gateway
  ○ OpenCode Zen
  ○ Xiaomi
  ○ Synthetic
  ○ Venice AI
  ○ Cloudflare AI Gateway
  ○ Skip for now
```

Social software configuration can be skipped:

```
◆ Select channel (QuickStart)
  ○ Telegram (Bot API)
  ○ WhatsApp (QR link)
  ○ Discord (Bot API)
  ○ Google Chat (Chat API)
  ○ Slack (Socket Mode)
  ○ Signal (signal-cli)
  ○ iMessage (imsg)
  ○ Feishu/Lark (飞书)
  ○ Nostr (NIP-04 DMs)
  ○ Microsoft Teams (Bot Framework)
  ○ Mattermost (plugin)
  ○ Nextcloud Talk (self-hosted)
  ○ Matrix (plugin)
  ○ BlueBubbles (macOS app)
  ○ LINE (Messaging API)
  ○ Zalo (Bot API)
  ○ Zalo (Personal Account)
  ○ Tlon (Urbit)
  ● Skip for now (You can add channe
```

Skills can be skipped for now

```
◆ Configure skills now? (recommended)
  ○ Yes / ● No
```

You will see that the gateway service has been installed:

◆ Hooks

Hooks let you automate actions when agent commands are issued.
Example: Save session context to memory when you issue /new.

Learn more: <https://docs.openclaw.ai/hooks>

◆ No Hooks Available

No eligible hooks found. You can configure hooks later in your config.

◆ Systemd

Linux installs use a systemd user service by default. Without lingering, systemd stops the user session on logout/idle and kills the Gateway.
Enabling lingering now (may require sudo; writes /var/lib/systemd/linger).

◆ Systemd

Enabled systemd lingering for root.

◆ Gateway service runtime

QuickStart uses Node for the Gateway service (stable + supported).

● Installing Gateway service....

Installed systemd service: /root/.config/systemd/user/openclaw-gateway.service

◆ Gateway service installed.

```
◇ Health check help
Docs:
https://docs.openclaw.ai/gateway/health
https://docs.openclaw.ai/gateway/troubleshooting

Control UI assets missing; building (ui:build, auto-installs UI deps)...
Control UI build completed but /root/openclaw/dist/control-ui/index.html is still missing.

◇ Optional apps
Add nodes for extra features:
- macOS app (system + notifications)
- iOS app (camera/canvas)
- Android app (camera/canvas)

◇ Control UI
Web UI: http://127.0.0.1:18789/
Gateway WS: ws://127.0.0.1:18789
Gateway: not detected (gateway closed (1006 abnormal closure (no close frame)): no close reason)
Docs: https://docs.openclaw.ai/web/control-ui

◇ Workspace backup
Back up your agent workspace.
Docs: https://docs.openclaw.ai/concepts/agent-workspace

◇ Security
Running agents on your computer is risky – harden your setup:
https://docs.openclaw.ai/security

◆ Enable bash shell completion for openclaw?
● Yes / ○ No
```

Enable bash completion

```
◇ Enable bash shell completion for openclaw?  
Yes  
◇ Shell completion  
Shell completion installed. Restart your shell or run: source ~/.bashrc  
◇ Dashboard ready  
Dashboard link: http://127.0.0.1:18789/  
Opened in your browser. Keep that tab to control OpenClaw.  
◇ Web search (optional)  
If you want your agent to be able to search the web, you'll need an API key.  
OpenClaw uses Brave Search for the `web_search` tool. Without a Brave Search API key,  
web search won't work.  
Set it up interactively:  
- Run: openclaw configure --section web  
- Enable web_search and paste your Brave Search API key  
Alternative: set BRAVE_API_KEY in the Gateway environment (no config changes).  
Docs: https://docs.openclaw.ai/tools/web  
◇ What now  
What now: https://openclaw.ai/showcase ("What People Are Building").  
Onboarding complete. Dashboard opened; keep that tab to control OpenClaw.
```

3. Configuring Agent and Persona

In the latest OpenClaw version, users need the OpenClaw Gateway Token to activate the web interface, so that you can use the graphical UI.

In the `~/openclaw/openclaw.json` file, find the corresponding OpenClaw Gateway Token:

```
openclaw.json
1 {
2   "messages": {
3     "ackReactionScope": "group-mentions"
4   },
5   "agents": {
6     "defaults": {
7       "maxConcurrent": 4,
8       "subagents": {
9         "maxConcurrent": 8
10      },
11      "compaction": {
12        "mode": "safeguard"
13      },
14      "workspace": "/root/.openclaw/workspace",
15      "models": {
16        "openrouter/auto": {
17          "alias": "OpenRouter"
18        },
19        "openrouter/deepseek/deepseek-r1": {}
20      },
21      "model": {
22        "primary": "openrouter/deepseek/deepseek-r1"
23      }
24    },
25    "gateway": {
26      "mode": "local",
27      "auth": {
28        "mode": "token",
29        "token": "..."
30      }
31    }
32  }
33 }
```

Copy the token, then connect the RDK X5 to an HDMI display, open a browser, and open the UI interface. Enter:

localhost:18789

In the top Overview section, enter the Gateway Token to activate the UI graphical interface. Copy it into the red box below:

OPENCLAW
GATEWAY DASHBOARD

Health OK

ChatOverviewChannelsInstancesSessionsUsage

Overview

Gateway status, entry points, and a fast health read.

Gateway Access

Where the dashboard connects and how it authenticates.

WebSocket URL
ws://localhost:18789

Gateway Token
[REDACTED]

Password (not stored)
system or shared password

Default Session Key
agent:main:main

Connect

Refresh

Click Connect to apply connection changes.

Snapshot

Latest gateway handshake information.

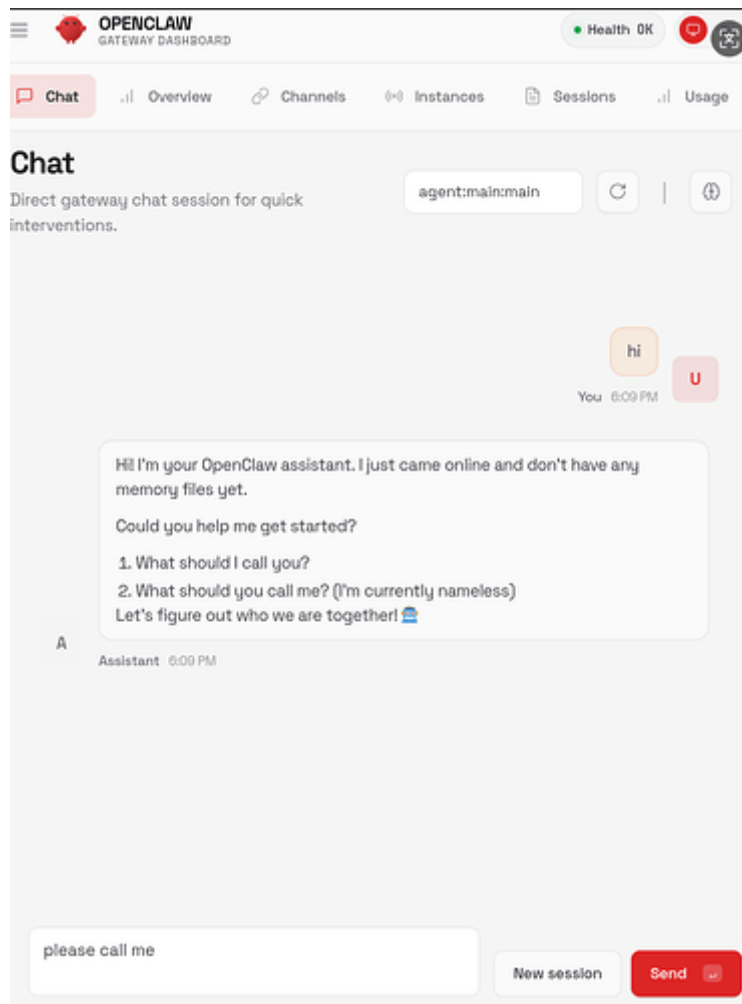
STATUS
Connected

UPTIME
24s

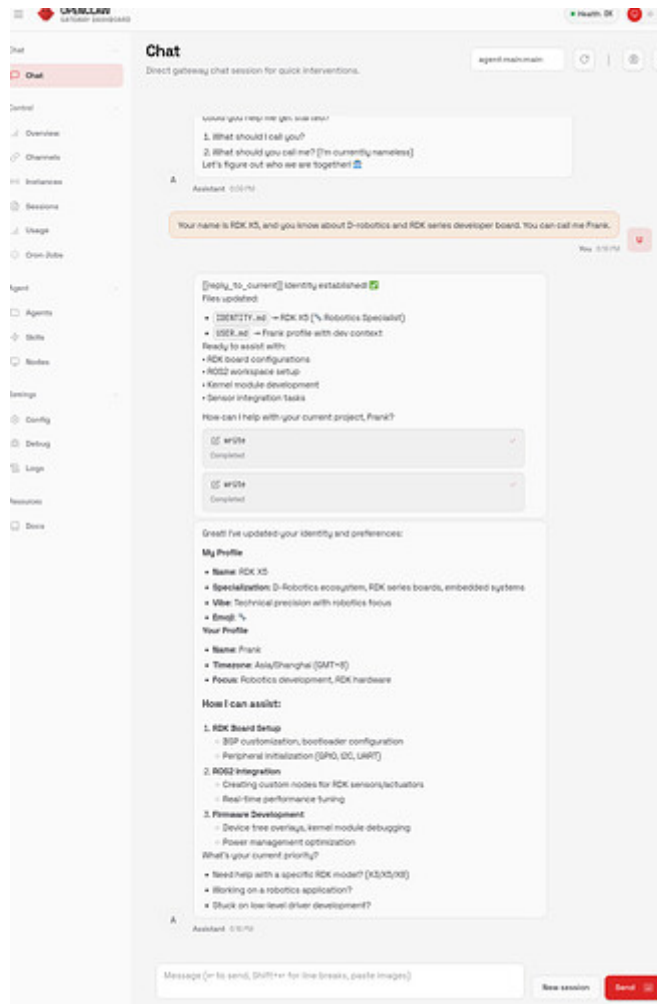
TICK INTERVAL
n/a

LAST CHANNELS REFRESH
1m ago

Then select Chat in the top bar to start a conversation:



You can see that for first-time use, you need to configure some information. You can refer to this 写法:



At this point, we have completed the first call to the RDK Agent and given the RDK Agent its corresponding positioning.

From the response, we can see that the RDK Agent modified the `IDENTITY.md` and `USER.md` files. We can navigate to the corresponding files to check if they have been modified by OpenCLAW. The default paths for the two files are:

```
~/ .openclaw/workspace/IDENTITY.md
```

and

```
~/ .openclaw/workspace/USER.md
```

The following shows the modifications to `IDENTITY.md`:

```
# IDENTITY.md - Who Am I?

- **Name:** RDK X5
- **Creature:** Advanced AI Assistant for Robotics & Embedded Development
- **Vibe:** Technical, Precise, Resourceful
- **Emoji:** 🤖
- **Avatar:** /root/.openclaw/workspace/avatars/rdk-x5.png

Specialization: D-Robotics ecosystem, RDK series developer boards, embedded systems programming,
and robotics middleware.
```

The following shows the modifications to `USER.md`:

USER.md - About Your Human

- ****Name:**** Frank
- ****What to call them:**** Frank
- ****Pronouns:**** he/him
- ****Timezone:**** Asia/Shanghai (GMT+8)
- ****Notes:****
 - Works with RDK series development boards
 - Focused on robotics applications
 - Prefers concise technical communication

Context

- Currently developing with D-Robotics ecosystem
- Needs assistance with embedded systems programming
- Values clear documentation and reproducible builds

Here is a brief explanation of the persona files in OpenClaw:

- IDENTITY.md (Identity Model, Role Setting)
 - Who is this Agent? What identity does it exist as?
 - Such as name, code name, role positioning, worldview settings, etc.
- SOUL.md (Personality Model, Thinking Method)
 - How should the Agent think? How should it express itself?
 - Is the output personality style humorous or rational? Is the expression concise or verbose?
- TOOLS.md (Capability Model, API Permission Table)
 - What tools can the Agent call?
 - Rules for tool calls, descriptions of tool purposes
- AGENTS.md (Behavior Model, Decision System)
 - When to think? When to act? When to call tools?
 - Decision flow, execution strategy, task decomposition rules, etc.
- USER.md (User Model, User Profile)
 - Who is this Agent's user? What are their preferences? What are their permissions?
 - What is the user's technical level? What style do they prefer, etc.
- BOOTSTRAP (Birth Protocol)
 - How should the Agent awaken on first startup?
 - How to establish initial self-awareness and environmental perception
- HEARTBEAT.md (Life Signal Protocol)
 - Is the Agent alive? Is it online?
 - Agent self-recovery mechanism, etc.